

Preet Navadiya

+91 9726222497 | preetnavadiya1609@gmail.com | Surat, Gujarat, India | linkedin.com/in/preet-navadiya-5794a7313

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

B.Tech, Electronics Engineering

CGPA: 7.43/10

2024 – 2028 (Expected)

Ashadeep IIT — HSC (2024) | 89% | 98.48 Percentile

RESEARCH INTERESTS

My research interests lie in Computer Vision, with a particular focus on low-light image enhancement, image restoration, and generative modeling. I am also developing a growing interest in Digital Electronics, CMOS, and VLSI design, and enjoy designing systems and architectures that achieve strong performance under real-world constraints.

I am particularly interested in:

- Low-Light Image Enhancement and Image Restoration for real-world degraded scenes
- Generative Adversarial Networks (GANs) for high-fidelity image synthesis
- Neural Machine Translation and Sequence-to-Sequence modeling with attention mechanisms
- Quantitative evaluation using PSNR, SSIM, and BLEU metrics for generative and translation tasks
- Digital Electronics, CMOS, and VLSI Design, including digital design flows and RTL-based system design

TECHNICAL SKILLS

- **Programming:** Python
- **Deep Learning:** TensorFlow, PyTorch
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV
- **Digital Electronics, CMOS & VLSI:** Digital Logic Design, CMOS Circuit Fundamentals, RTL Design, Verilog/SystemVerilog basics, Digital Design Flows, EDA Tool Ecosystems, Simulation-based Verification
- **Tools & Environments:** Git, Linux, Jupyter Notebook, Google Colab, VS Code
- **Evaluation Metrics:** PSNR, SSIM (image quality), BLEU (translation quality)

PROJECTS & RESEARCH

NTIRE 2026 — Low-Light Image Enhancement

(Ongoing)

- Developed a deep learning enhancement model targeting the NTIRE 2026 competitive image restoration benchmark.
- Achieved PSNR of 29.97 dB and SSIM of 0.8726 on the evaluation dataset.
- Systematically optimized model architecture and training pipeline using quantitative perceptual metrics.
- Conducting ablation studies to improve robustness under diverse and challenging lighting conditions.

English–Hindi Neural Machine Translation

- Designed and implemented a Seq2Seq encoder–decoder architecture with Bahdanau attention mechanism.
- Trained on a parallel English–Hindi corpus and evaluated translation quality using BLEU score.
- Analyzed attention weight distributions to interpret model focus across source sentence tokens.

GAN-based Image Generator

- Implemented a full adversarial training pipeline including generator and discriminator networks from scratch.
- Improved training stability and output quality through systematic hyperparameter tuning and loss scheduling.

Machine Learning Prediction System

- Built a supervised regression model achieving 88% accuracy on the test set.
- Applied feature engineering, missing value imputation, and normalization preprocessing pipelines.

TECHNICAL PARTICIPATION

NTIRE 2026 — New Trends in Image Restoration and Enhancement (CVPR Workshop)

(Ongoing)

Participating in the NTIRE 2026 Computer Vision Challenge on low-light image enhancement, developing and benchmarking deep learning models evaluated against international participants on PSNR and SSIM metrics.

DVCon 2026

(Ongoing)

Attended the Design and Verification Conference 2026, gaining exposure to advanced semiconductor design methodologies, digital electronics, CMOS, and VLSI verification workflows, and EDA tool ecosystems.

CERTIFICATION

Tata EDA Certification — Completed industry-recognized certification covering Electronic Design Automation fundamentals, digital design flows, and simulation-based verification techniques relevant to CMOS and VLSI design.